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Effect of Plant Extract of *Parthenium* on Seed Germination and Seedling Growth of *Zea mays* L.

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Abstract : The present investigation deals with the effect of plant extract of *Parthenium* on seed gernimation and seedling growth of two varieties of *Zea mays* L..The diffreent treatment were based on different doses of *Parthenium* extract. At lower dose the germination percentage was high. The germination % and Relative Germination Ratio decreases as we increases the dose of the extract. The root and shoot length increases in treatment T_1 and T_2 and then decreases. The Var. DHM-103 was more sensitive than Var. GS-2.

Key Words: *Zea mays* L., *Parthenium*, Germination %, Relative Germination Ratio.

INTRODUCTION

Cereals are the most important sources of plant food for mankind. *Zea mays* L. commonly called as Maize, Corn, Makai and Bhutta belongs to family Poaceae. Maize producing countries are U.S.A., India, China, Russia, Manchuria, Yogoslavia and Mexico. Maize contain very high percentage of cabohydrates(68% - 70%), a considerable amount of proteins, some fats and vitamins. The plant is said to have lots of medicinal properties.

Parthenium (Asteraceae) commonly called as congress grass, carrot grass and stare weed is native of Tropical America belived that, it entered the Indian soil in between 1950-56 along with wheat sent by U.S.A.. The word " Parthenium" is derived from Latin word " Parthenice" suggesting medical use.

MATERIALS AND METHODS

The seeds of two varieties of *Zea mays* L. that is Var. DHM-103 and Var. GS-2 were collected from B.A.U., Kanke and Seed nursery, Ranchi. The seeds of Var. DHM-103 were yellow and the seeds of Var. GS-2 were white. The biological mutagen selected was plant extract of *Parthenium.* Plant extract was prepared by grinding the green leaves with inflorescence in water and filtering it. Five different treatment which were based on different doses of *Parthenium extract* were used for seed treatment were as follows:-.

T_0 = Seeds treated with distilled water only (control)
T_1 = Seeds treated with 5% concentrations of *Parthenium* extract
T_2 = Seeds treated with 10% concentrations of *Parthenium* extract
T_3 = Seeds treated with 15% concentrations of *Parthenium* extract
T_4 = Seeds treated with 20% concentrations of *Parthenium* extract
T_5 = Seeds treated with 25% concentrations of *Parthenium* extract

About 600 seeds were dipped in water for imbibition. After two hour the seeds were taken off and 100 seeds were soaked in each treatment for 10 hours. Then the seeds were washed in water and wrapped in cotton cloth for germination. The number of seed germinated were counted and the length of root and shoot of seedlings were measured. Relative Elongation Ratio(RER)of and shoot were also calculated [1.]

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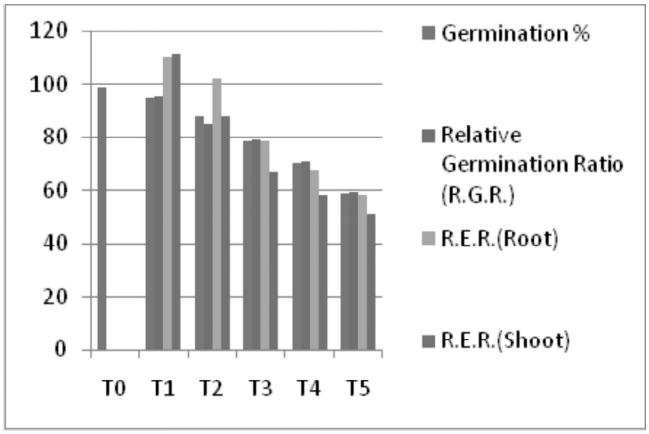
RESULT AND DISCUSSION

The data obtained for seed germination and seedling growth were presented in table- 1 and fig.-1. The germination % was recorded 99.33% in both the varieties of *Zea mays* L. in treatment T_0. The germination % decreases as we increase the concentration of *Parthenium* extract in Var. DHM-103 and Var. GS-2 but Var. DHM-103 showed much lower germination as compared to Var. GS-2. This might be due to sensivity of Var. DHM-103[2,3]. On increasing the concentration of *Parthenium* extract a gradual decrease in germination % and relative germination ratio were reported which is in accordance with the result obtained earlier[4,5,6]. In treatment T_0 the root length in var. DHM-103 was 6.35cm and in Var. GS-2 it was 5.06cm. There was marked increase in the root length of both the varieties in treatment in T_1 and T_2 and then it decreases as we increase the concentration. The shoot length also increases after treatment but decreases at certain concentration of plant extract. The growth of root and shoot length in both the varieties showed an expoential relationship with the dose [7,8]. In the growth behaviour of root length the Var. DHM-103 was more sensitive than var. GS-2 whereas in the growth of shoot length GS-2 is more sensitive. While considering the seed germination and growth behaviour the var. DHM-103 was more sensitive towards different concentration of *Parthenium* extract.

Table-1 : Germination and Seedling Growth in *zea mays L.* var. DHM – 103 and var. GS – 2 after treatment with Plant extract of *Parthenium*

Varieties	Treatment	Germination %	Relative Germination Ratio (R.G.R.)	Root Length	Relative Elongation Ratio of Root (R.E.R.)	Shoot Length	Relative Elongation Ratio of Shoot (R.E.R.)
Var. DHM – 103	T_0	99.33 ± 0.66	–	6.35 ± 0.17	_	4.13 ± 0.07	_
	T_1	95.00 ± 3.46	95.64	7.04 ± 0.02	110.86	4.62 ± 0.01	111.86
	T_2	88.00 ± 4.05	85.59	6.50 ± 0.02	102.36	3.64 ± 0.00	88.13
	T_3	78.66 ± 2.30	79.19	5.00 ± 0.01	78.74	2.78 ± 0.01	67.31
	T_4	70.33± 0.66	70.80	4.31 ± 0.02	67.87	2.42 ± 0.01	58.59
	T_5	59.00± 2.30	59.39	3.72 ± 0.17	58.58	2.12 ± 0.01	51.33
Var. GS – 2	T_0	99.33 ± 0.33	_	5.06 ± 0.02	_	3.07 ± 0.04	–
	T_1	98.66 ± 0.33	99.32	5.92 ± 0.01	116.99	4.26 ± 0.00	138.76
	T_2	94.33 ± 0.88	94.96	5,61 ± 0.02	110.86	4.14 ± 0.02	134.85
	T_3	87.66 ± 0.88	88.25	4.82 ± 0.02	95.25	3.31 ± 0.03	107.81
	T_4	78.66± 0.66	79.19	4.74 ± 0.02	93.67	2.83 ± 0.01	92.18
	T_5	70.00± 0.57	70.47	3.34 ± 0.02	66.00	2.44 ± 0.02	79.47

Var. DHM – 103



Var. GS-2

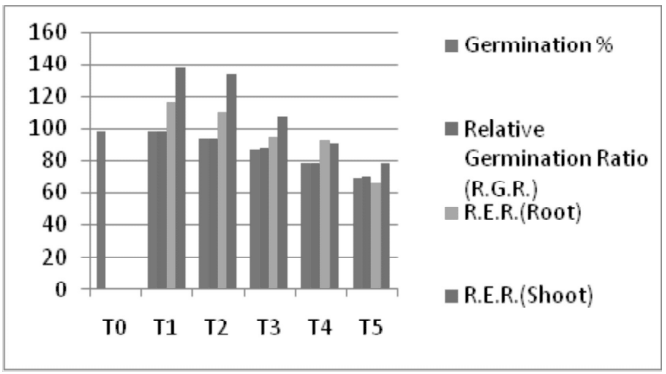


Fig. 1: Column graph showing Germination and Seedling Growth in *zea mays L.* Var. DHM – 103 and Var. GS – 2 after treatment with Plant extract of *Parthenium*

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